



SHRI RAMSWAROOP MEMORIAL UNIVERSITY LUCKNOW



LAB MANUAL *Session 2026-2027*

Database Management System Lab (UCS5208)

Course :B.Tech.

Branch :CSE Year:III Semester:Vth

Name of Lab Faculty: Dr. Yusuf Perwej



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B.Tech. [SEM V] (CS)

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Objective 1 :- Write the queries for Data Definition and Data Manipulation Language.

Perform DDL and DML operations to create, modify, and manipulate database objects and validate solutions using HackerRank Basic SQL Challenges.

THEORY:

Data Definition Language (DDL): Used to define and modify database schema.

- CREATE - Create database objects (tables, indexes, views)
- ALTER - Modify existing database objects
- DROP - Delete database objects
- TRUNCATE - Remove all records from a table
- RENAME - Rename database objects

Data Manipulation Language (DML): Used to manipulate data in tables.

- INSERT - Add new rows to a table
- UPDATE - Modify existing data
- DELETE - Remove rows from a table
- SELECT - Retrieve data from tables

APPARATUS / SOFTWARE REQUIRED:

S.No.	Item	Version	Quantity
1	MySQL Server / Oracle / PostgreSQL	8.0+ / 19c+	1
2	MySQL Workbench / SQL Developer	Latest	1
3	HackerRank Account	-	1
4	Operating System	Windows/Linux	1

**PROCEDURE:****PART A: DDL OPERATIONS**

1. Create Database:

sql

```
CREATE DATABASE UniversityDB;
```

```
USE UniversityDB;
```

2. Create Table:

sql

```
CREATE TABLE Students (  
    StudentID INT PRIMARY KEY,  
    FirstName VARCHAR(50) NOT NULL,  
    LastName VARCHAR(50) NOT NULL,  
    Email VARCHAR(100) UNIQUE,  
    DateOfBirth DATE,  
    EnrollmentDate DATE DEFAULT CURRENT_DATE,  
    GPA DECIMAL(3,2) CHECK (GPA >= 0.0 AND GPA <= 4.0)  
);
```

```
CREATE TABLE Courses (  
    CourseID INT PRIMARY KEY,  
    CourseName VARCHAR(100) NOT NULL,  
    Credits INT CHECK (Credits >= 1 AND Credits <= 5),  
    Department VARCHAR(50)  
);
```

```
CREATE TABLE Enrollments (  
    EnrollmentID INT PRIMARY KEY,  
    StudentID INT,  
    CourseID INT,  
    Grade CHAR(2),  
    EnrollmentDate DATE DEFAULT CURRENT_DATE,  
    FOREIGN KEY (StudentID) REFERENCES Students(StudentID),  
    FOREIGN KEY (CourseID) REFERENCES Courses(CourseID)  
);
```

3. Alter Table Operations:

sql

```
-- Add a new column
```



```
ALTER TABLE Students ADD COLUMN PhoneNumber VARCHAR(15);
```

-- *Modify column datatype*

```
ALTER TABLE Students MODIFY COLUMN PhoneNumber VARCHAR(20);
```

-- *Rename column*

```
ALTER TABLE Students RENAME COLUMN PhoneNumber TO ContactNumber;
```

-- *Drop column (use with caution)*

```
ALTER TABLE Students DROP COLUMN ContactNumber;
```

4. Drop and Truncate:

```
sql
```

-- *Remove all rows but keep structure*

```
TRUNCATE TABLE Enrollments;
```

-- *Delete entire table*

```
DROP TABLE IF EXISTS Enrollments;
```

PART B: DML OPERATIONS

5. Insert Data:

```
sql
```

-- *Insert single record*

```
INSERT INTO Students (StudentID, FirstName, LastName, Email, DateOfBirth, GPA)  
VALUES (101, 'John', 'Doe', 'john.doe@email.com', '2000-05-15', 3.5);
```

```
INSERT INTO Students VALUES
```

```
(102, 'Jane', 'Smith', 'jane.smith@email.com', '1999-08-22', '2024-09-01', 3.8);
```

-- *Insert multiple records*

```
INSERT INTO Students (StudentID, FirstName, LastName, Email, DateOfBirth, GPA) VALUES  
(103, 'Michael', 'Johnson', 'michael.j@email.com', '2001-03-10', 3.2),  
(104, 'Emily', 'Brown', 'emily.b@email.com', '2000-11-28', 3.9),  
(105, 'David', 'Wilson', 'david.w@email.com', '1999-07-19', 2.8);
```

6. Update Data:

```
sql
```

-- *Update specific record*

```
UPDATE Students
```



SET GPA = 3.7

WHERE StudentID = 101;

-- *Update multiple columns*

UPDATE Students

SET GPA = 3.1, Email = 'david.wilson@email.com'

WHERE StudentID = 105;

-- *Update all records (use with caution)*

UPDATE Students

SET GPA = GPA + 0.1

WHERE GPA < 3.0;

7. Delete Data:

sql

-- *Delete specific record*

DELETE FROM Students

WHERE StudentID = 105;

-- *Delete with condition*

DELETE FROM Students

WHERE GPA < 2.0;

-- *Delete all records (use with caution)*

DELETE FROM Students;

8. Select Data:

sql

-- *Select all columns*

SELECT * FROM Students;

-- *Select specific columns*

SELECT FirstName, LastName, Email FROM Students;

-- *Select with condition*

SELECT * FROM Students

WHERE GPA >= 3.5;

-- *Select with ordering*



```
SELECT * FROM Students
ORDER BY LastName ASC, FirstName ASC;
```

HACKERRANK CHALLENGES VALIDATION:**Challenge 1: Revising the Select Query I**

```
sql
```

-- Problem: Query all columns for all American cities in CITY table with populations larger than 100000

```
SELECT *
FROM CITY
WHERE COUNTRYCODE = 'USA'
AND POPULATION > 100000;
```

Challenge 2: Employee Salaries

```
sql
```

-- Problem: Write a query that prints a list of employee names whose salary > 2000 and months < 10

```
SELECT name
FROM Employee
WHERE salary > 2000
AND months < 10
ORDER BY employee_id;
```

Challenge 3: Weather Observation Station 1

```
sql
```

-- Problem: Query the list of CITY and STATE from STATION table

```
SELECT CITY, STATE
FROM STATION;
```

OBSERVATION TABLE:

Operation	SQL Command
Create Database	CREATE DATABASE UniversityDB;
Create Tables	CREATE TABLE Students ...
Alter Table	ALTER TABLE Students ADD ...
Insert Data	INSERT INTO Students VALUES ...
Update Data	UPDATE Students SET ...



Operation	SQL Command
Delete Data	DELETE FROM Students WHERE ...
Select Data	SELECT * FROM Students ...
Truncate	TRUNCATE TABLE Enrollments;
Drop Table	DROP TABLE Enrollments;